INCH-POUND

MIL-C-24643/38D
22 November 1994
SUPERSEDING
MIL-C-24643/38C
14 March 1994

MILITARY SPECIFICATION SHEET

# CABLE, ELECTRICAL, 600 VOLTS, TYPE LSECM (INCLUDING VARIATION LSECMA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

This cable can be manufactured in two variations: LSECM (unarmored) and LSECMA (armored).

REQUIREMENTS:

Qualification required.

## Construction (watertight)

First - Copper conductor, in accordance with ASTM B 286, size 20-7, tin coated. Maximum diameter 0.039 inch.

Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.

Third - Crosslinked polyethylene insulation. Nominal wall thickness 0.023 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements. Colored insulation, one black and one white conductor for each pair.

* Fourth - Two conductors with a nominal lay of 2.5 inches to form each a pair.

Fifth - Binder tape over each pair, applied at manufacturer's option, applied helically with overlap.

* Sixth - Braided shield of AWS No. 34 or No. 36 tin coated copper. Braid Angle of 30 to 35 degrees; minimum coverage of 85 percent

Seventh - Shield insulation of two polyester tapes, type G in accordance with MIL-I-631, outer tape sealed, An alternate shield insulation may consist of one tape, type G of MIL-I-631, plus an extruded jacket of a suitable transparent material, minimum average thickness of the transparent material 0.003 inch. The standard identification code shall be applied by method 2 on the inner tape.


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Construction (watertight) (continued)

Groups of 7 (8 groups)

First - Copper conductor, AWG no. 18, class B stranding, (7 by 0.0152 inch) uncoated. Maximum diameter 0.050 inch.
Second - Ethylene propylene rubber or cross-linked polyethylene insulation, white. Nominal wall thickness 0.008 inch. Colored insulation five black and two white conductors for each group.
Third - Seven conductors, five black and two white, with the white conductors adjacent in the layer of six, cabled with a lay not grater than 2 inches, to form each group.
Fourth - Two polyester tapes, type G in accordance with MIL-I-631, outer tape sealed, An alternate shield insulation may consist of one tape, type G of MIL-I-631, plus an extruded jacket of a suitable transparent material, minimum average thickness of the transparent material 0.003 inch. The standard identification code shall be applied by method 2 on the inner tape.

ASSEMBLY:

First - Five pairs cable to form a core. Three pairs and eight groups of seven cabled over the core, with a lay not greater than 24 times the pitch diameter of the layer. Pair nos. 1 through 5 in core; the sequence of the outer layer - pair no. 6, groups nos. 1, 2 and 3; pair no. 7; groups nos. 4, 5 and 6; pair no. 8; group nos. 7 and 8. Fillers shall be used as necessary to form a well rounded assembly.
Second - Binder tape applied helically with overlap.

Construction variant for type LSECM

* Third - Cross-linked polyolefin jacket. Cable jacket thickness 0.075 inch minimum average.

Construction variant for type LSECMA

Third - Cross-linked polyolefin jacket. Cable jacket thickness 0.075 inch minimum average.
Fourth - Braided metal armor.

NOTE: 1. Variant LSECM completed cable diameters: Minimum 1.290 inches, maximum 1.370 inches. Military part number M24643/38-01UN.
2. Variant LSECMA completed cable diameters: Minimum 1.340 inches, maximum 1.420 inches. Military part number M24643/38-02AN.



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| EXAMINATION AND TESTS: | Requirements |
|---|---|
| Basic electrical: | |
| Conductor Resistance - ohms/1000 feet at 25°C, maximum | |
| ASTM B 286, size 20-7 | 11.03 |
| AWG no 18 Class B | 7.47 |
| Voltage withstand - volts, root mean square, minimum | |
| Conductor to conductor | 2000 |
| Shield to shield | 500 |
| Insulation resistance - Megohms/1000 feet minimum | |
| Conductor to shield | 500 |
| Component shield to shield | 100 |
| Conductor and shield continuity | No failure |
| Group A: | |
| Visual and dimensional | No failure |
| Watertightness - see MIL-C-24643 for limits of water leakage | No failure |
| Capacitance | |
| Mutual capacitance - 1 kilohertz (KHz), pF/ft, maximum | 30 |
| Group B: | |
| Cross-linked proof test (percent, maximum) | |
| Insulation | 50 |
| Jacket (When tested at 200°C) | 50 |
| Tear - pounds per inch thickness, minimum (ASTM D 470) | 35 |
| Physicals (unaged) | |
| Insulation | |
| Ethylene propylene rubber | |
| Tensile strength - $lb/in^2$, minimum | 700 |
| Elongation - percent, minimum | 250 |
| Cross-linked polyethylene | |
| Tensile strength - $lb/in^2$, minimum | 1800 |
| Elongation - percent, minimum | 250 |
| Jacket (cable) | |
| Tensile strength - $lb/in^2$, minimum | 1300 |
| Elongation - percent, minimum | 160 |
| Group C: | |
| Physicals (aged) | |
| Insulation | |
| Ethylene propylene | |
| Air oven | |
| Tensile strength - percent of unaged, minimum | 75 |
| Elongation - percent of unaged minimum | 75 |


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| EXAMINATION AND TESTS (continued): | Requirements |
|---|---|
| Cross-linked polyethylene | |
| Air oven | |
| Tensile strength - percent of unaged, minimum | 80 |
| Elongation - percent of unaged minimum | 80 |
| Jacket (cable) | |
| Air oven | |
| Tensile strength - percent of unaged, minimum | 60 |
| Elongation - percent of unaged minimum | 60 |
| Hot oil immersion | |
| Tensile strength - percent of unaged, minimum | 50 |
| Elongation - percent of unaged minimum | 50 |
| Shrinkage | No failure |
| Heat distortion - percent of unaged, maximum | 30 |
| * Permanence of printing (jacket) - cycles, minimum | 125 |
| Cable sealant removability | No failure |
| * Armor (type LSECMA only) - conformance to material, construction and coverage | No failure |
| Group D: | |
| Flame propagation (cable) | No failure |

| QUALIFICATION INSPECTION: | Requirements |
|---|---|
| Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following: | |
| Aging and compatibility (cable) (125± 5°C) | No failure |
| Abrasion resistance (jacket) - scrapes, minimum | 75 |
| Acid gas equivalent - percent, maximum | |
| Jacket | 2 |
| Fillers | 2 |
| Insulation | 18 |
| Shield insulation group covering | 2 |
| Halogen content - percent, maximum | |
| Jacket | 0.2 |
| Fillers | 0.2 |
| Immersion (jacket) | |
| Tensile strength - percent of unaged, minimum | 50 |
| Elongation - percent of unaged, minimum | 50 |
| Smoke index, maximum | |
| Jacket | 25 |
| Fillers | 45 |
| Insulation | 45 |



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QUALIFICATION INSPECTION:

| | Requirements |
|---|---|
| Toxicity index, maximum | |
| Jacket | 5 |
| Fillers | 5 |
| Insulation | 1.5 |
| Durometer (jacket) - (type A) hardness, minimum | 80 |
| Weathering (jacket) | No failure |

UNIT ORDERING LENGTHS:
1000 feet nominal

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Preparing Activity:
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Agent:
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(Project No. 6145-2101-38)



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